

MO7019: Statistical Methods for Climate Science

Chap. 1

Exercise 1

If $X = \{x_1, \dots, x_n\}$ with $\Pr(X = x_i) = p_i$; $p_1 + \dots + p_n = 1$; $p_i \geq 0$.

$$E(X) = \sum_{i=1}^n x_i p_i, \text{ and } \text{var}(X) = E(X^2) - (E(X))^2 = \sum x_i^2 p_i - \left(\sum x_i p_i\right)^2.$$

For the Bernoulli random variable:

$$E(X) = 0 \cdot p + 1 \cdot q = q = 1 - p$$

$$\text{var}(X) = q^2 - q = q(1 - q) = pq.$$

Exercise 2

For the Poisson distribution with $P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$

$$E(X) = \sum k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum \frac{\lambda^k}{(k-1)!} e^{-\lambda} = \lambda \sum \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} = \lambda \Rightarrow E(X) = \lambda$$

$$E(X^2) = \left(\sum k^2 \frac{\lambda^k}{k!} \right) e^{-\lambda} = e^{-\lambda} \sum k \cdot \frac{\lambda^k}{(k-1)!} = \lambda e^{-\lambda} \left[\sum k \cdot \frac{\lambda^{k-1}}{(k-1)!} \right] = \lambda e^{-\lambda} f(\lambda)$$

integrating $f(\lambda) \Rightarrow \int f(\lambda) = \sum \frac{\lambda^k}{(k-1)!} = \lambda \sum \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{\lambda} \Rightarrow f(\lambda) = (1+\lambda)e^{\lambda}$

$$\Rightarrow E(X^2) = \lambda(1+\lambda) \Rightarrow \text{var}(X) = \lambda$$

Exercise 3

we keep in mind: $\frac{1}{\sqrt{2\pi}} \int e^{-x^2/2} dx = 1$.

• $E(X) = \frac{1}{\sigma\sqrt{2\pi}} \int x e^{-(x-m)/2\sigma^2} dx \rightarrow$ variable change $y = \frac{x-m}{\sigma} \Rightarrow E(X) = \frac{\sigma}{\sqrt{2\pi}} \int y e^{-y^2/2} dy + \frac{m}{\sqrt{2\pi}} \int e^{-y^2/2} dy = m$

• Variance - assume $m=0$. we start from $\frac{1}{\sigma\sqrt{2\pi}} \int e^{-x^2/2\sigma^2} dx = 1$ + integrate by part:

$$\sigma\sqrt{2\pi} = \int \underbrace{1}_{u'} \cdot \underbrace{e^{-x^2/2\sigma^2}}_v dx = x e^{-x^2/2\sigma^2} \Big|_{-\infty}^{\infty} - \int x \left(\frac{-2x}{2\sigma^2} \right) e^{-x^2/2\sigma^2} dx = \frac{1}{\sigma^2} \int x^2 e^{-x^2/2\sigma^2} dx = \sigma\sqrt{2\pi}$$

$$\text{so } \frac{1}{\sigma\sqrt{2\pi}} \int x^2 e^{-x^2/2\sigma^2} dx = \sigma^2 = \text{var}(X).$$

• Skewness: $E(X^3) = 0$; symmetric distribution.

• Kurtosis - $E(X^4) = \frac{1}{\sigma\sqrt{2\pi}} \int x^4 e^{-x^2/2\sigma^2} dx$. We start from $\frac{1}{\sigma\sqrt{2\pi}} \int x^2 e^{-x^2/2\sigma^2} dx$

and integrate by part.

$$\sigma^2 = \frac{1}{\sigma\sqrt{2\pi}} \int \underbrace{x^2}_{u'} \cdot \underbrace{e^{-x^2/2\sigma^2}}_v dx = \frac{1}{\sigma\sqrt{2\pi}} \left[\frac{x^3}{3} e^{-x^2/2\sigma^2} \right]_{-\infty}^{\infty} - \frac{1}{\sigma\sqrt{2\pi}} \int \frac{x^3}{3} \left(\frac{-2x}{2\sigma^2} \right) e^{-x^2/2\sigma^2} dx$$

$$\text{so } \sigma^2 = \frac{1}{3\sigma^2} \cdot \frac{1}{\sigma\sqrt{2\pi}} \int x^4 e^{-x^2/2\sigma^2} dx \Rightarrow E(X^4) = 3\sigma^4$$

• This can be extended to $E(X^k)$, $k \geq 5$, etc.

Chap. 2

Exercise 1

$X_{t+1} = \alpha X_t + \varepsilon_{t+1}$, ε_t : independent and identically distributed (iid) variables with mean zero and unit variance.

$$\Rightarrow X_t \cdot X_{t+1} = \alpha X_t^2 + X_t \varepsilon_{t+1}$$

take expectation: $E(X_t X_{t+1}) = \alpha E(X_t^2) + E(X_t \varepsilon_{t+1})$;

assuming $E(X_t) = 0$ and $\text{var}(X_t) = 1 \Rightarrow \alpha = \frac{E(X_t X_{t+1})}{E(X_t^2)} = \rho(1)$.

$\rho(1)$: autocorrelation function of $X_t, t \geq 1$.

Next, take the square of the equation: $X_{t+1}^2 = \alpha^2 X_t^2 + \varepsilon_{t+1}^2 + 2\alpha X_t \varepsilon_{t+1}$ and take expectation:

$$E(X_{t+1}^2) = \alpha^2 E(X_t^2) + \sigma^2 \Rightarrow \underline{\sigma^2 = 1 - \alpha^2 = 1 - \rho(1)^2}$$

Exercise 2

For a AR(2): $X_{t+2} = \alpha_1 X_{t+1} + \alpha_2 X_t + \varepsilon_{t+2}$

Apply the same procedure: multiply by X_{t+1} , and also by X_t and take expectation:

$$\begin{cases} \rho(1) = \alpha_1 + \alpha_2 \rho(1) \\ \rho(2) = \alpha_1 \rho(1) + \alpha_2 \end{cases} \rightarrow \begin{bmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \rho(1) \\ \rho(2) \end{bmatrix}$$

$$\text{So, } \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = T^{-1} \begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix}, \text{ where } T = \begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix}$$

$$\alpha_1 = \frac{1 - \rho(2)}{1 - \rho(1)^2} \cdot \rho(1), \quad \alpha_2 = 1 - \frac{1 - \rho(2)}{1 - \rho(1)^2}$$

For a AR(3): $X_{t+3} = \alpha_1 X_{t+2} + \alpha_2 X_{t+1} + \alpha_3 X_t + \varepsilon_{t+3}$, and we get:

$$\begin{pmatrix} 1 & \rho(1) & \rho(2) \\ \rho(1) & 1 & \rho(1) \\ \rho(2) & \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \rho(3) \end{pmatrix}$$

For a AR(m) model: $X_{t+m} = \alpha_1 X_{t+m-1} + \dots + \alpha_m X_t + \varepsilon_{t+m}$

$$\begin{pmatrix} 1 & \rho(1) & \dots & \rho(m-1) \\ \rho(1) & 1 & & \\ \vdots & & \ddots & \\ \rho(m-1) & \dots & \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(m) \end{pmatrix}$$

Chap. 3

Exercise

We consider $\alpha = 5\%$ significance level.

- For the time series x_t : $H_0: \mu_x = 0$ versus $H_1: \mu_x \neq 0$

Let σ be the (unknown) standard deviation of x_t , based on the null H_0 ,

$$Z = \frac{\bar{x} - 0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

$$\frac{\bar{x}}{s/\sqrt{n}} = \frac{\bar{x}/\sigma\sqrt{n}}{s/\sigma} = \frac{Z}{\sqrt{\frac{U}{n-1}}} = T_{n-1}; \quad \left(U = \frac{\sum x_t^2}{\sigma^2} \sim \chi_{n-1}^2 \right)$$

Here $n-1=5$,

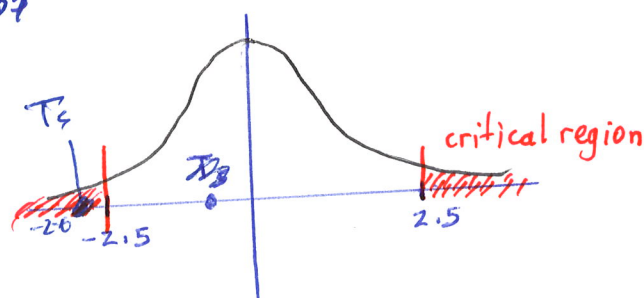
$$T_5 = \frac{\bar{x}}{s/\sqrt{n}} = \frac{-1.2 \times \sqrt{6}}{1.1} = -2.67$$

The $\frac{\alpha}{2}$ 'th quantile of the student distribution with 5 dof is

$$t_5(0.025) = -2.5; \text{ so}$$

$$|T_5| > |t_5(0.025)|$$

Conclusion: reject the null hypothesis H_0 .



- For the y_t time series — we find:

$$\frac{\bar{y}}{s/\sqrt{n-1}} = \frac{0.63 \sqrt{5}}{0.52} = 2.7 = T_5$$

$$|T_5| > |t_5(0.025)| \rightarrow \text{Conclusion: reject } H_0.$$

- Now, we want to compare μ_x and μ_y : $H_0: \mu_x = \mu_y$ versus $H_1: \mu_x \neq \mu_y$.

We can assume x_t and y_t have the same variance σ^2 (unknown).

then we can consider the statistic:

$$T = \frac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; \quad s^2 = \text{pooled variance};$$

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

$$n_1 = n_2 = 6; \quad s_1 = s_x; \quad s_2 = s_y; \quad T \sim t_{n_1+n_2-2}$$

$$\rightarrow T = \frac{-1.2 - 0.63}{0.94/\sqrt{3}} = -3.46; \quad \text{and } t_{\frac{\alpha}{2}, 10} = -1.8 \rightarrow |T| > |t_{\frac{\alpha}{2}, 10}| \rightarrow \text{reject } H_0$$

we can also use the case when σ_1 and σ_2 are both unknown: $T = \frac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$

$$T \sim t_{\min(n_1-1, n_2-1)} = t_5 \rightarrow T = -3.7; \quad \text{and } t_{\alpha/2, 5} = -2.01 \rightarrow \text{reject } H_0.$$

Chap. 5

Exercise 1

$$y_t = \alpha + \beta x_t + \varepsilon_t \quad ; \quad t=1, \dots, n \quad ; \quad E(\varepsilon_t) = 0.$$

We can assume $\bar{x}_t = \bar{y}_t = 0$, and take $\alpha = 0$.

$$\text{From } y_t = \beta x_t + \varepsilon_t \rightarrow \hat{\beta} = \arg \min \left[\sum_{t=1}^n (y_t - \beta x_t)^2 \right]$$

$$\text{we obtain } \hat{\beta} = \frac{\sum x_t y_t}{\sum x_t^2} ; \quad \text{Denote } S_{xx} = \sum x_t^2 ; S_{xy} = \sum x_t y_t$$

$$\text{and } S_{yy} = \sum y_t^2 = SST$$

$$E[\hat{\beta}] = \frac{\sum x_t E(y_t)}{\sum x_t^2} = \frac{\sum x_t \beta x_t}{\sum x_t^2} = \beta$$

$\hat{\beta}$: is an unbiased estimator of β .

$$E[\hat{\beta}^2] = \frac{E(\sum x_t y_t)^2}{(\sum x_t^2)^2} = \frac{\sum_{t,s} x_t x_s E(y_t y_s)}{(\sum x_t^2)^2} = \frac{\sum_{t,s} x_t x_s E(\beta^2 x_t x_s + \sum \varepsilon_t \varepsilon_s + \beta x_t \varepsilon_s + \beta \varepsilon_t x_s)}{(\sum x_t^2)^2}$$

$$= \frac{\sum_{t,s} x_t x_s \beta^2 x_t x_s + \sum x_t x_s \sigma^2 \delta_{ts}}{(\sum x_t^2)^2} ; \quad \delta_{ts} = \begin{cases} 1 & \text{if } t=s \\ 0 & \text{if } t \neq s \end{cases}$$

$$= \beta^2 + \frac{\sigma^2}{\sum x_t^2}$$

$$\text{so } \text{var}(\hat{\beta}) = E(\hat{\beta}^2) - (E(\hat{\beta}))^2 = \frac{\sigma^2}{\sum x_t^2} = \frac{\sigma^2}{S_{xx}}$$

$$\text{so } \hat{\beta} \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{S_{xx}}\right)$$

Exercise 2

Assuming again $\bar{x} = \bar{y} = 0$.

$$R^2 = \frac{SSR}{SST} = \frac{\hat{\beta} S_{xy}}{S_{yy}} = \frac{\sum x_t y_t}{\sum x_t^2} \cdot \frac{\sum x_t y_t}{\sum y_t^2} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r_{xy}^2$$

$$\text{since } r_{xy} = \frac{\text{cov}(x_t, y_t)}{\sqrt{\text{var}(x_t) \text{var}(y_t)}} = \frac{\frac{1}{n} \sum x_t y_t}{\sqrt{\frac{1}{n} \sum x_t^2 \cdot \frac{1}{n} \sum y_t^2}} = \frac{\sum x_t y_t}{\sqrt{\sum x_t^2 \sum y_t^2}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

Exercise 3

1. One can take the mean out for all time series.

$$pna = a \cdot nino3 + \epsilon, \quad a = 0.41.$$

In Matlab: $[B \text{ BINT } R \text{ RINT } STATS] = \text{regress}(pna, [\text{ones}(12), nino3])$;

$$STATS = \begin{matrix} 0.33 & 5.07 & 0.048 & 0.39 \\ R^2 & F & p & \sigma_{\epsilon}^2 \end{matrix}$$

$p = 0.048$ - The regression coefficient is significant at 5% (but not at 1% level).

Note that $\text{corr}(pna, nino3) = 0.6$

2. Multiple regression -

$$nao = b \cdot pna + c \cdot nino3 + \epsilon$$

$$b = -0.09; \quad c = -0.2$$

$$STATS = \begin{matrix} 0.06 & 0.28 & 0.76 & 1.22 \\ R^2 & F & p & \sigma_{\epsilon}^2 \end{matrix}$$

Note that $\text{corr}(nao, pna) = -0.18$; $\text{corr}(nao, nino3) = -0.23$.

3. The predictors pna and $nino3$ are not independent - they are correlated. One way out is to seek a linear combination to make non-correlated variables. - Can use PCA:

$X = [pna \ nino3]$; 12×2 array. then apply SVD:

$$[U \ S \ V] = \text{svd}(X);$$

explained variance: ~~68~~ 68% and 32% - (for PC1 and PC2).

We can then fit the model: $nao = \alpha \cdot PC1 + \epsilon$; $\alpha = -0.82$

$$STATS = \begin{matrix} 0.06 & 0.61 & 0.45 & 1.1 \end{matrix}$$

→ Note the reduction of the p-value. (but not substantially).

Q. What happens if we use both PCs in the regression.

A. We get same result as in question 2. above, since all data will the whole variance is used (no information is lost).

Chap. 6

Exercise 1

The covariance between the i 'th and j 'th variables (or grid points) is:

$$s_{ij} = \frac{1}{n-1} \sum_{t=1}^n x_{t,i} x_{t,j}$$

Now, the i 'th and j 'th element of $\frac{1}{n-1} X^T X$ is the sum of the product between the i 'th and j 'th columns of X , i.e.,

$$\frac{1}{n-1} \begin{pmatrix} x_{1,i} \\ \vdots \\ x_{n,i} \end{pmatrix} \cdot \begin{pmatrix} x_{1,j} \\ \vdots \\ x_{n,j} \end{pmatrix} = \frac{1}{n-1} \sum_{t=1}^n x_{t,i} x_{t,j} = s_{ij}$$

Exercise 2

At time t , the projection of the data $x_t = \begin{pmatrix} x_{t,1} \\ \vdots \\ x_{t,p} \end{pmatrix}$ onto the vector $a = \begin{pmatrix} a_1 \\ \vdots \\ a_p \end{pmatrix}$ is $a^T \cdot x_t = \sum_{i=1}^p a_i x_{t,i} = y_t$; $t=1, \dots, n$

we obtain a time series y_t , $t=1, \dots, n$. The mean of y_t , $t=1, \dots, n$, is zero : $\bar{y} = 0$.

The variance of y_t is

$$\begin{aligned} \text{var}(y) &= \frac{1}{n-1} \sum_{t=1}^n y_t^2 = \frac{1}{n-1} \sum_{t=1}^n \left[\sum_{i=1}^p \sum_{j=1}^p a_i a_j x_{t,i} x_{t,j} \right] \\ &= \sum_{i,j=1}^p a_i a_j \left(\frac{1}{n-1} \sum_{t=1}^n x_{t,i} x_{t,j} \right) = \sum_{i,j=1}^p a_i a_j s_{ij} = a^T S a. \end{aligned}$$

Maximizing this variance : $\max(a^T S a)$:

To make the problem bounded, we assume $\|a\|=1$; $(a^T a = 1)$:

$$\begin{cases} \max a^T S a \\ \text{s.t. } a^T a - 1 = 0 \end{cases}$$

Transform the problem to : $\max_a [a^T S a - \lambda (a^T a - 1)] = \max_a [f(a)]$

$f(a) = a^T S a - \lambda (a^T a - 1)$; λ : Lagrange multiplier ; $\frac{\partial f}{\partial a} = 2Sa - 2\lambda a = 0$.

$\Rightarrow Sa = \lambda a$; So a is an eigenvector of S .

Now, $a^T S a = \lambda a^T a = \lambda \Rightarrow a$ is the eigenvector of S with the largest eigenvalue $\lambda \Rightarrow$ leading eigenvector. (Note $\lambda > 0$).